Molecular Characterization Laboratory
Frederick National Laboratory for Cancer Research
(Leidos Biomedical Research, Inc.)
459 Miller Drive, Frederick, MD 21702
T: 301.846.7689

Patient Name:		Patient DOB:	
Date Collected:		Referring Physician:	
Biopsy Site:	**Lymph Node**	Primary Diagnosis:	**Metastatic Melanoma**
Sample ID:		MoCha ID:	
Telephone:		Fax:	
Tumor Content (%):	**70%**		

Tumor content assessment performed by Van Andel Research Institute; Grand Rapids; MI 49503

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Report Highlights

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Relevant Melanoma Findings

Gene	Finding
BRAF	***BRAF V600K, BRAF amplification***
KIT	Not detected
NTRK1	Not detected
NTRK2	Not detected
NTRK3	Not detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	*BRAF V600K*	**atezolizumab + cobimetinib + vemurafenib** [1] **binimetinib + encorafenib** [1] **cobimetinib + vemurafenib** [1] **dabrafenib + trametinib** [1] **trametinib** [1] dabrafenib vemurafenib	dabrafenib vemurafenib	53
IIC	*MET amplification*	None	capmatinib crizotinib	9
III	*PTEN V133_C136del*	None	None	18

Public data sources included in relevant therapies: FDA1, NCCN
Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Relevant Biomarkers (continued)

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IIC	*CDK6 amplification*	None	None	10
IIC	*BRAF amplification*	None	None	8
III	*ATRX K425Q*	None	None	8

Public data sources included in relevant therapies: FDA1, NCCN
Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
BRAF	p.(V600K)	c.1798_1799delGTinsAA	COSM473	chr7:140453136	76.22%	NM_004333.4	missense
PTEN	p.(V133_C136del)	c.398_409delTAATGATATGTG	.	chr10:89692910	35.60%	NM_000314.4	nonframeshift Deletion
ATRX	p.(K425Q)	c.1273A>C	.	chrX:76939475	44.17%	NM_000489.4	missense

Copy Number Variations

Gene	Locus	Copy Number
CDK6	chr7:92244371	5.23
MET	chr7:116339591	5.41
BRAF	chr7:140439604	4.71

Comments:

CNVs are slightly below the clinical reporting threshold for this assay.

Signed By: ______________________________

Biomarker Descriptions

ATRX (ATRX, chromatin remodeler)

Background: The ATRX gene encodes the ATRX chromatin remodeler and ATPase/helicase domain protein which belongs to SWI/SNF family of chromatin remodeling proteins[1]. The SWI/SNF proteins are a group of DNA translocases that use ATP-hydrolysis to remodel chromatin structure and maintain genomic integrity by controlling transcriptional regulation, DNA repair, and chromosome stability through its regulation of telomere length[2,3,4,5]. ATRX is a tumor suppressor that interacts with the MRE11-RAD50-NBN (MRN) complex which is involved in double-stranded DNA (dsDNA) break repair[6,7,8].

Alterations and prevalence: Somatic mutations of ATRX are observed in 36% of diffuse glioma, 14% of sarcoma, 13% of endometrial cancer, and 8% of melanoma cases[9,10].

Potential relevance: Currently, no therapies are approved for ATRX aberrations. Loss of ATRX protein expression correlates with the presence of ATRX mutations[11,12]. The loss of ATRX expression along with mutations in IDH1 and p53 is a diagnostic criteria for astrocytoma as defined by the World Health Organization(WHO)[11].

BRAF (B-Raf proto-oncogene, serine/threonine kinase)

Background: The BRAF gene encodes the B-Raf proto-oncogene serine/threonine kinase, a member of the RAF family of serine/threonine protein kinases which also includes ARAF and RAF1 (CRAF). BRAF is among the most commonly mutated kinases in cancer. Activation of the MAPK pathway occurs through BRAF mutations and leads to an increase in cell division, dedifferentiation, and survival[13,14]. BRAF mutations are categorized into three distinct functional classes namely, class 1, 2, and 3, and are defined by the dependency on the RAS pathway. Class 1 and 2 BRAF mutants are RAS-independent in that they signal as active monomers (Class 1) or dimers (Class 2) and become uncoupled from RAS GTPase signaling, resulting in constitutive activation of BRAF[15]. Class 3 mutants are RAS dependent as the kinase domain function is impaired or dead[15,16,17].

Alterations and prevalence: Recurrent somatic mutations in BRAF are observed in 40-60% of melanoma and thyroid cancer, approximately 10% of colorectal cancer, and about 2% of non-small cell lung cancer (NSCLC)[9,10,18,19,20]. Mutations at V600 belong to class 1 and include V600E, the most recurrent somatic BRAF mutation across diverse cancer types[16,21]. Class 2 mutations include K601E/N/T, L597Q/V, G469A/V/R/, G464V/E/, and BRAF fusions[16]. Class 3 mutations include D287H, V459L, G466V/E/A, S467L, G469E, and N581S/I[16]. BRAF V600E is universally present in hairy cell leukemia, mature B-cell cancer, and prevalent in histiocytic neoplasms[22,23,24]. Other recurrent BRAF somatic mutations cluster in the glycine-rich phosphate-binding loop at codons 464-469 in exon 11 as well as additional codons flanking V600 in the activation loop[21]. In primary cancers, BRAF amplification is observed in 8% of ovarian cancer and about 1% of breast cancer[9,10]. BRAF fusions are mutually exclusive to BRAF V600 mutations and have been described in melanoma, thyroid cancer, pilocytic astrocytoma, NSCLC, and several other cancer types[25,26,27,28,29]. Part of the oncogenic mechanism of BRAF gene fusions is the removal of the N-terminal auto-inhibitory domain leading to constitutive kinase activation[17,25,27].

Potential relevance: Vemurafenib[30] (2011) was the first targeted therapy approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E mutation. BRAF class 1 mutations, including V600E, are sensitive to vemurafenib, whereas class 2 and 3 mutations are insensitive[16]. BRAF kinase inhibitors including dabrafenib[31] (2013) and encorafenib[32] (2018) are also approved for the treatment of patients with unresectable or metastatic melanoma with BRAF V600E/K mutations. Encorafenib[32] is approved in combination with cetuximab (2020) for the treatment of BRAF V600E mutated colorectal cancer. Due to the tight coupling of RAF and MEK signaling, several MEK inhibitors have been approved for patients harboring BRAF alterations[16]. Trametinib[33] (2013) and binimetinib[34] (2018) were approved for the treatment of metastatic melanoma with BRAF V600E/K mutations. Combination therapies of BRAF plus MEK inhibitors have been approved in melanoma and NSCLC. The combinations of dabrafenib/trametinib (2015) and vemurafenib/cobimetinib[35] (2015) were approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E mutation. Subsequently, the combination of dabrafenib and trametinib was approved for metastatic NSCLC (2017) with a BRAF V600E mutation. The pan-RAF kinase inhibitor DAY-101 was granted breakthrough therapy designation (2020) by the FDA for pediatric patients with advanced low-grade glioma harboring activating RAF alterations[36]. The ERK inhibitor ulixertinib[37] was also granted a fast track designation in 2020 for the treatment of patients with non-colorectal solid tumors harboring BRAF mutations G469A/V, L485W, or L597Q. BRAF fusion is a suggested mechanism of resistance to BRAF targeted therapy in melanoma[38]. Additional mechanisms of resistance to BRAF targeted therapy include BRAF amplification and alternative splice transcripts as well as activation of PI3K signaling and activating mutations in KRAS, NRAS, and MAP2K1/2 (MEK1/2)[39,40,41,42,43,44,45]. Clinical responses to sorafenib and trametinib in limited case studies of patients with BRAF fusions have been reported[29].

Biomarker Descriptions (continued)

CDK6 (cyclin dependent kinase 6)

Background: The CDK6 gene encodes the cyclin dependent kinase 6 protein, a homologue of CDK4. Both proteins are serine/threonine protein kinases and involved in the regulation of the G1/S phase transition of the mitotic cell cycle[46,47]. Following complex formation of CDK6 with D-type cyclins (e.g., CCND1, CCND2 or CCND3), CDK6 kinase activation leads to the phosphorylation of retinoblastoma protein (RB) followed by E2F activation, DNA replication, and cell cycle progression[48].

Alterations and prevalence: Recurrent somatic mutations of CDK6 have not been characterized. CDK6 is recurrently amplified in esophageal carcinoma (10-15%), stomach adenocarcinoma (5-10%), lung squamous cell carcinoma (5%), and head and neck squamous cell carcinoma (4%)[9,10,49,50].

Potential relevance: Currently, no therapies are approved for CDK6 aberrations. Small molecule inhibitors targeting CDK4/6 including palbociclib (2015), abemaciclib (2017), and ribociclib (2017) are FDA approved in combination with an aromatase inhibitor or fulvestrant for the treatment of hormone receptor positive, HER2-negative advanced or metastatic breast cancer.

MET (MET proto-oncogene, receptor tyrosine kinase)

Background: The MET proto-oncogene encodes a receptor tyrosine kinase for the hepatocyte growth factor (HGF) protein, which is expressed by mesenchymal cells. Ubiquitin-dependent proteolysis regulates the steady state level of the MET protein via recognition of the tyrosine phosphorylation site Y1003 in the MET Cbl-binding domain within the juxtamembrane region[51,52,53]. Growth factor signaling leads to MET dimerization and subsequent initiation of downstream effectors including those involved in the RAS/RAF/MEK/ERK and PI3K/AKT signaling pathways, which regulate cell migration, proliferation, and survival[54,55].

Alterations and prevalence: Recurrent somatic MET alterations include activating mutations, gene amplification, and translocations generating MET gene fusions. Recurrent somatic mutations fall into two classes, mutations in the MET kinase domain, which are uncommon, and splice-site mutations affecting exon 14. Recurrent kinase domain mutations are observed in papillary renal cell carcinoma (PRCC) (1-2%) and include M1250T, H1094Y, and V1070E. Mutation of the Y1003 phosphorylation site is reported in lung cancer but is uncommon (<1%)[9,56]. In contrast, splice-site mutations flanking exon 14 are observed in 4% of non-small cell lung cancer (NSCLC). These mutations include canonical splice site mutations affecting exon 14 and deletions that extend into the splicing motifs within intron 13[57,58]. Such mutations disrupt splicing leading to the formation of an alternative transcript that joins exon 13 directly to exon 15 and skips exon 14 entirely. The MET exon 14 skipping transcript lacks the juxtamembrane domain that contains the recognition motif for ubiquitin-dependent proteolysis and thus leads to a marked increase in steady-state level of the MET protein[59]. MET exon 14 skipping mutations act as oncogenic drivers in lung cancer mutually exclusive to activating mutations in EGFR and KRAS and other oncogenic fusions such as ALK and ROS1[57,60,61]. MET is amplified in 2-5% of ovarian cancer, esophageal adenocarcinoma, stomach adenocarcinoma, glioblastoma, and lung adenocarcinoma[9,49,56]. Recurrent MET fusions, although infrequent, are observed in adult and pediatric glioblastoma, papillary renal cell carcinoma, lung cancer, liver cancer, thyroid cancer, and melanoma[62,63,64]. MET alterations are believed to be enriched in late-stage cancers where they drive tumor progression and metastasis[65,66,67].

Potential relevance: In 2020, the FDA granted accelerated approval to capmatinib[68] for NSCLC harboring MET exon 14 skipping positive as detected by an FDA-approved test[69]. Tepotinib[70] has been granted FDA breakthrough designation (2019) for advanced MET exon 14 skipping NSCLC. MET exon 14 skipping mutations confer sensitivity to approved kinase inhibitors including crizotinib (2011), which is recommended for MET amplifications and exon 14 skipping mutations[57,60,61,69]. Conversely, amplification of MET has been observed to mediate resistance to EGFR tyrosine kinase inhibitors (TKIs)[71,72,73,74,75]. In a phase II trial testing the MET inhibitor savolitinib, patients with advanced PRCC exhibited median progression free survival (PFS) of 6.2 and 1.4 months for MET-driven and MET-independent PRCC, respectively[76].

PTEN (phosphatase and tensin homolog)

Background: The PTEN gene encodes the phosphatase and tensin homolog, a tumor suppressor protein with lipid and protein phosphatase activities[77]. PTEN antagonizes PI3K/AKT signaling by catalyzing the dephosphorylation of phosphatidylinositol (3,4,5)-trisphosphate (PIP3) to PIP2 at the cell membrane, which inhibits the activation of AKT[78,79]. Germline mutations in PTEN are linked to hamartoma tumor syndromes, including Cowden disease, which are defined by uncontrolled cell growth and benign or malignant tumor formation[80]. PTEN germline mutations are also associated with inherited cancer risk in several cancer types[81].

Alterations and prevalence: PTEN is frequently altered in cancer by inactivating loss-of-function mutations and by gene deletion. PTEN mutations are frequently observed in 50%-60% of uterine cancer[9,10]. Nearly half of somatic mutations in PTEN are stop-gain or frame-

Biomarker Descriptions (continued)

shift mutations that result in truncation of the protein reading frame. Recurrent missense or stop-gain mutations at codons R130, R173, and R233 result in loss of phosphatase activity and inhibition of wild-type PTEN[79,82,83,84,85]. PTEN gene deletion is observed in 15% of prostate cancer, 9% of squamous lung cancer, 9% of glioblastoma, and 1-5% of melanoma, sarcoma, and ovarian cancer[9,10].

Potential relevance: Currently, no therapies are approved for PTEN aberrations. However, due to the role of PTEN in genome stability, poly(ADP-ribose) polymerase inhibitors (PARPi) are being explored as a potential therapeutic strategy in PTEN deficient tumors[86,87].

Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

BRAF V600K

Relevant Therapy	FDA	NCCN	Clinical Trials*
cobimetinib + vemurafenib	●	●	● (II)
binimetinib + encorafenib	●	●	✖
dabrafenib + trametinib	●	●	✖
atezolizumab + cobimetinib + vemurafenib	●	✖	✖
trametinib	●	✖	✖
vemurafenib	✖	◐	● (II)
dabrafenib	✖	◐	✖
bempegaldesleukin, nivolumab	✖	✖	● (III)
dabrafenib, trametinib, ipilimumab, nivolumab	✖	✖	● (III)
relatlimab, nivolumab	✖	✖	● (II/III)
sargramostim, nivolumab, ipilimumab	✖	✖	● (II/III)
abemaciclib + LY3214996	✖	✖	● (II)
atezolizumab, cobimetinib, vemurafenib	✖	✖	● (II)
binimetinib, encorafenib	✖	✖	● (II)
buparlisib	✖	✖	● (II)
cobimetinib, atezolizumab, vemurafenib	✖	✖	● (II)
cobimetinib, vemurafenib	✖	✖	● (II)
dabrafenib, nivolumab, trametinib	✖	✖	● (II)
dabrafenib, pembrolizumab, trametinib	✖	✖	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

BRAF V600K (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
dabrafenib, trametinib	✖	✖	● (II)
dabrafenib, trametinib, spartalizumab	✖	✖	● (II)
encorafenib, binimetinib, nivolumab, ipilimumab	✖	✖	● (II)
IMM-101, nivolumab, ipilimumab	✖	✖	● (II)
immunostimulant, pembrolizumab	✖	✖	● (II)
ipilimumab, binimetinib, nivolumab, encorafenib	✖	✖	● (II)
ipilimumab, nivolumab, encorafenib, binimetinib, dabrafenib, trametinib	✖	✖	● (II)
LXH254 , LTT-462, trametinib, ribociclib	✖	✖	● (II)
nivolumab, ipilimumab, radiation therapy	✖	✖	● (II)
ulixertinib	✖	✖	● (II)
vemurafenib, cobimetinib	✖	✖	● (II)
vemurafenib, cobimetinib, atezolizumab	✖	✖	● (II)
vemurafenib, cobimetinib, ipilimumab, nivolumab	✖	✖	● (II)
vemurafenib, cobimetinib, surgical intervention, radiation therapy	✖	✖	● (II)
vemurafenib, dabrafenib	✖	✖	● (II)
ASTX029	✖	✖	● (I/II)
bemcentinib, dabrafenib, pembrolizumab, trametinib	✖	✖	● (I/II)
dabrafenib, trametinib, antimalarial	✖	✖	● (I/II)
dabrafenib, trametinib, navitoclax	✖	✖	● (I/II)
HH-2710	✖	✖	● (I/II)
mirdametinib, lifirafenib	✖	✖	● (I/II)
ABM-1310	✖	✖	● (I)
AZD-0364	✖	✖	● (I)
BGB-3245	✖	✖	● (I)
CART-GD2, vemurafenib	✖	✖	● (I)
cobimetinib, belvarafenib	✖	✖	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

BRAF V600K (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
DAY-101	✖	✖	● (I)
E6201	✖	✖	● (I)
HL-085, vemurafenib	✖	✖	● (I)
itacitinib, dabrafenib, trametinib	✖	✖	● (I)
JAB-3312	✖	✖	● (I)
JSI-1187, dabrafenib	✖	✖	● (I)
LY3214996, midazolam, abemaciclib, chemotherapy, encorafenib, cetuximab	✖	✖	● (I)
RG-7461, pembrolizumab	✖	✖	● (I)
RMC-4630	✖	✖	● (I)
RO-5126766, everolimus	✖	✖	● (I)

MET amplification

Relevant Therapy	FDA	NCCN	Clinical Trials*
crizotinib	✖	○	● (II)
capmatinib	✖	○	✖
cabozantinib	✖	✖	● (II)
tepotinib	✖	✖	● (II)
GST-HG161	✖	✖	● (I)
HLX55	✖	✖	● (I)
HS-10241	✖	✖	● (I)

PTEN V133_C136del

Relevant Therapy	FDA	NCCN	Clinical Trials*
berzosertib	✖	✖	● (II)
ipatasertib	✖	✖	● (II)
niraparib	✖	✖	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

PTEN V133_C136del (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
olaparib	✖	✖	● (II)
paxalisib	✖	✖	● (II)
samotolisib	✖	✖	● (II)
talazoparib	✖	✖	● (II)
temsirolimus	✖	✖	● (II)
copanlisib, nivolumab, ipilimumab	✖	✖	● (I/II)
ipatasertib, atezolizumab	✖	✖	● (I/II)
TAS-117, futibatinib	✖	✖	● (I/II)
copanlisib, olaparib, durvalumab	✖	✖	● (I)
gedatolisib + palbociclib	✖	✖	● (I)
HWH-340	✖	✖	● (I)
paxalisib, radiation therapy	✖	✖	● (I)

CDK6 amplification

Relevant Therapy	FDA	NCCN	Clinical Trials*
abemaciclib	✖	✖	● (II)
palbociclib	✖	✖	● (II)
palbociclib, abemaciclib	✖	✖	● (II)
siremadlin, ribociclib	✖	✖	● (II)
SY-5609	✖	✖	● (I)

BRAF amplification

Relevant Therapy	FDA	NCCN	Clinical Trials*
regorafenib	✖	✖	● (II)
ulixertinib	✖	✖	● (II)
ASTX029	✖	✖	● (I/II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

BRAF amplification (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
mirdametinib, lifirafenib	✖	✖	● (I/II)
AZD-0364	✖	✖	● (I)
BGB-3245	✖	✖	● (I)
LY3214996, midazolam, abemaciclib, chemotherapy, encorafenib, cetuximab	✖	✖	● (I)
RMC-4630	✖	✖	● (I)

ATRX K425Q

Relevant Therapy	FDA	NCCN	Clinical Trials*
atezolizumab	✖	✖	● (II)
niraparib	✖	✖	● (II)
nivolumab, talazoparib	✖	✖	● (II)
olaparib	✖	✖	● (II)
BAY-1895344, niraparib	✖	✖	● (I)
copanlisib, olaparib, durvalumab	✖	✖	● (I)
VX-803	✖	✖	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

BRAF V600K

NCT ID	Title	Phase
NCT03898908	Phase II, Multicentre Clinical Trial to Evaluate the Activity of Encorafenib and Binimetinib Administered Before Local Treatment in Patients With BRAF Mutant Melanoma Metastatic to the Brain.	II
NCT02858921	A Phase II, Randomised, Open Label Study of Neoadjuvant Dabrafenib, Trametinib and / or Pembrolizumab in BRAF V600 Mutant Resectable Stage IIIB/C Melanoma	II
NCT02231775	Neoadjuvant and Adjuvant Dabrafenib and Trametinib in Patients With Clinical Stage III Melanoma (Combi-Neo)	II
NCT04310397	Altering Adjuvant Therapy Based on Pathologic Response to Neoadjuvant Dabrafenib and Trametinib (ALTER-PATH NeoDT)	II

Clinical Trials Summary (continued)

BRAF V600K (continued)

NCT ID	Title	Phase
NCT03235245	Combination of Targeted Therapy (Encorafenib and Binimetinib) Followed by Combination of Immunotherapy (Ipilimumab and Nivolumab) vs Immediate Combination of Immunotherapy in Patients With Unresectable or Metastatic Melanoma With BRAF V600 Mutation : an EORTC Randomized Phase II Study (EBIN)	II
NCT02968303	Phase II Study With COmbination of Vemurafenib With Cobimetinib in B-RAF V600E/K Mutated Melanoma Patients to Normalize LDH and Optimize Nivolumab and Ipilimumab therapy	II
NCT01989585	Phase I/II Study of Dabrafenib, Trametinib, and Navitoclax in BRAF Mutant Melanoma (Phase I and II) and Other Solid Tumors (Phase I Only)	I/II
No NCT ID	Phase I Study of Safety and Immune Effects of an Escalating Dose of Autologous GD2 Chimeric Antigen Receptor-Expressing Peripheral Blood T cells in Patients with Metastatic Melanoma	I
NCT03272464	Phase I Study of INCB039110 in Combination With Dabrafenib and Trametinib in Patients With BRAF-mutant Melanoma and Other Solid Tumors	I
NCT02224781	DREAMseq (Doublet, Randomized Evaluation in Advanced Melanoma Sequencing) a Phase III Trial	III
NCT03554083	Neoadjuvant Therapy for Patients With High Risk Stage III Melanoma: A Pilot Clinical Trial	II
NCT03911869	A Phase II, Open-Label, Randomized, Multicenter Trial of Encorafenib + Binimetinib Evaluating a Standard-dose and a High-dose Regimen in Patients With BRAFV600-mutant Melanoma Brain Metastasis	II
NCT02452294	An Open-label, Uncontrolled, Single Arm Phase II Trial of Buparlisib in Patients With Metastatic Melanoma With Brain Metastases Not Eligible for Surgery or Radiosurgery	II
NCT03625141	A Phase II Two Cohort Study Evaluating the Safety and Efficacy of Cobimetinib Plus Atezolizumab in BRAFV600 Wild-type Melanoma With Central Nervous System Metastases and Cobimetinib Plus Atezolizumab and Vemurafenib in BRAFV600 Mutation-positive Melanoma With Central Nervous System Metastases.	II
NCT03430947	An Open-Label Phase II Multicenter Study Of Vemurafenib (Zelboraf) Plus Cobimetinib (Cotellic) After Radiosurgery In Patients With Active BRAF-V600-Mutant Melanoma Brain Metastases	II
NCT04511013	A Randomized Phase II Trial of Encorafenib + Binimetinib + Nivolumab vs Ipilimumab + Nivolumab in BRAF-V600 Mutant Melanoma With Brain Metastases	II
NCT04417621	A Randomized, Open-label, Multi-arm, Two-part, Phase II Study to Assess Efficacy and Safety of Multiple LXH254 Combinations in Patients With Previously Treated Unresectable or Metastatic BRAFV600 or NRAS Mutant Melanoma	II
NCT03224208	VECODUE A Phase II Trial of Vemurafenib Plus Cobimetinib in Patients Treated With Prior First-line Systemic Immunotherapy for Inoperable Locally Advanced or Metastatic Melanoma	II
NCT02303951	Neoadjuvant Treatment With the Combination of Vemurafenib, Cobimetinib and Atezolizumab in Limited Metastasis of Malignant Melanoma (AJCC Stage IIIC/IV) and Integrated Biomarker Study: a Single Armed, Two Cohort, Phase II EADO Trial NEO-VC	II
NCT03514901	An Evaluation of the Efficacy Beyond Progression of Vemurafenib Combined With Cobimetinib Associated With Local Treatment Compared to Second-line Treatment in Patients With BRAFV600 Mutation-positive Metastatic Melanoma in Focal Progression With First-line Combined Vemurafenib and Cobimetinib.	II

Clinical Trials Summary (continued)

BRAF V600K (continued)

NCT ID	Title	Phase
NCT03754179	A lead-in Phase I Followed by a Phase II Clinical Trial on the Combination of Dabrafenib, Trametinib and the Autophagy Inhibitor Hydroxychloroquine in BRAF/MEK Inhibitor-pretreated Patients With Advanced BRAF V600 Mutant Melanoma	I/II
NCT04190628	A Phase I, First-In-Human, Multicenter, Open-Label Study of ABM-1310, Administered Orally in Adult Patients With Advanced Solid Tumors	I
NCT04249843	A First-in-Human, Phase Ia/Ib, Open Label, Dose-Escalation and Expansion Study to Investigate the Safety, Pharmacokinetics, and Antitumor Activity of the RAF Dimer Inhibitor BGB-3245 in Patients With Advanced or Refractory Tumors	I
NCT03284502	A Phase Ib, Open-label, Multicenter, Dose Escalation Study of the Safety, Tolerability, and Pharmacokinetics of Cobimetinib and HM95573 in Patients With Locally Advanced or Metastatic Solid Tumors	I
NCT04418167	A Phase I Study of ERK1/2 Inhibitor JSI-1187 Administered as Monotherapy and in Combination With Dabrafenib for the Treatment of Advanced Solid Tumors With MAPK Pathway Mutations	I
NCT03635983	A Phase III, Randomized, Open-label Study of NKTR-214 Combined With Nivolumab Versus Nivolumab in Participants With Previously Untreated Unresectable or Metastatic Melanoma.	III
NCT03470922	A Randomized, Double-Blind Phase II/III Study of Relatlimab Combined With Nivolumab Versus Nivolumab in Participants With Previously Untreated Metastatic or Unresectable Melanoma	II/III
NCT03711188	A Study of the Safety and Efficacy of IMM-101 in Combination With Checkpoint Inhibitor Therapy in Patients With Advanced Melanoma	II
NCT02339571	Randomized Phase II/III Study of Nivolumab Plus Ipilimumab Plus Sargramostim Versus Nivolumab Plus Ipilimumab in Patients With Unresectable Stage III or Stage IV Melanoma	II/III
NCT02910700	A Phase II Study of the TRIplet Combination of Dabrafenib, Nivolumab, and Trametinib in Patients With Metastatic Melanoma (TRIDeNT)	II
NCT03563729	Efficacy Of Immunotherapy In Melanoma Patients With Brain Metastases Treated With Steroids	II
NCT02872259	A Phase Ib/II Randomised Open Label Study of BGB324 in Combination With Pembrolizumab or Dabrafenib/Trametinib Compared to Pembrolizumab or Dabrafenib/Trametinib Alone, in Patients With Advanced Non-resectable (Stage IIIc) or Metastatic (Stage IV) Melanoma.	I/II
NCT03332589	A Phase I Study Of E6201 For The Treatment Of Central Nervous System Metastases (CNS) From BRAF Or MEK-Mutated Metastatic Melanoma	I
NCT04079166	A Phase II Combination Study of SCIB1 with Pembrolizumab in Patients with Stage III or Stage IV Metastatic Melanoma	II
NCT03340129	A Phase II, Open Label, Single Arm Study of Ipilimumab and Nivolumab With Salvage Radiotherapy in Patients With Melanoma Brain Metastases	II
NCT03875079	An Open-Label, Multicenter, Phase Ib Study To Evaluate Safety And Therapeutic Activity Of RO6874281, An Immunocytokine, Consisting Of Interleukin-2 Variant (IL-2v) Targeting Fibroblast Activation Protein-? (FAP), In Combination With Pembrolizumab (Anti-PD-1), In Participants With Advanced Or Metastatic Melanoma	I

Clinical Trials Summary (continued)

BRAF V600K (continued)

NCT ID	Title	Phase
NCT03905148	A Phase Ib, Open-Label, Dose-escalation and Expansion Study to Investigate the Safety, Pharmacokinetics and Antitumor Activities of a RAF Dimer Inhibitor BGB-283 in Combination With MEK Inhibitor PD-0325901 in Patients With Advanced or Refractory Solid Tumors	I/II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT03155620	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice) Screening Protocol	II
NCT03220035	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase II Subprotocol of Vemurafenib in Patients With Tumors Harboring Braf V600 Mutations	II
NCT03781219	A Phase I, Single Arm, Dose Escalation Study to Evaluate Safety, Pharmacokinetics and Preliminary Efficacy of HL-085 Plus Vemurafenib in Patients With BRAF V600 Mutant Advanced Solid Tumor	I
NCT04534283	A Phase II Basket Trial of an ERK1/2 Inhibitor (LY3214996) in Combination With Abemaciclib for Patients Whose Tumors Harbor Pathogenic Alterations in BRAF, RAF1, MEK1/2, ERK1/2, and NF1	II
NCT03239015	Efficacy and Safety of Targeted Precision Therapy in Refractory Tumor With Druggable Molecular Event	II
NCT04045496	A Phase I, Multi-Center, Open-Label Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Preliminary Evidence of Antitumor Activity of JAB-3312 in Adult Patients With Advanced Solid Tumors	I
NCT02407509	A Phase I Trial of RO5126766 (a Dual RAF/MEK Inhibitor) Exploring Intermittent, Oral Dosing Regimens in Patients With Solid Tumours or Multiple Myeloma, With an Expansion to Explore Intermittent Dosing in Combination With Everolimus	I
NCT04488003	A Two-Part, Phase II, Multi-center Study of the ERK Inhibitor Ulixertinib (BVD-523) for Patients With Advanced Malignancies Harboring MEK or Atypical BRAF Alterations	II
NCT04198818	A First-in-Human, Open Label, Phase I/II Study to Evaluate the Safety, Tolerability and Pharmacokinetics of HH2710 in Patients With Advanced Tumors	I/II
NCT03429803	A Phase I Study of TAK-580 (MLN2480) for Children With Low-Grade Gliomas and Other RAS/RAF/MEK/ERK Pathway Activated Tumors	I
NCT03634982	A Phase I, Open-Label, Multicenter, Dose-Escalation Study of RMC-4630 Monotherapy in Adult Participants with Relapsed/Refractory Solid Tumors	I
NCT03520075	A Phase I/II Study of the Safety, Pharmacokinetics, and Activity of ASTX029 in Subjects With Advanced Solid Tumors	I/II
NCT04305249	A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of ATG-017 Monotherapy in Patients With Advanced Solid Tumors and Hematological Malignancies	I
NCT02857270	A Phase I Study of an ERK1/2 Inhibitor (LY3214996) Administered Alone or in Combination With Other Agents in Advanced Cancer	I

Clinical Trials Summary (continued)

MET amplification

NCT ID	Title	Phase
NCT04116541	MegaMOST - A Multicenter, Open-label, Biology Driven, Phase II Study Evaluating the Activity of Anti-cancer Treatments Targeting Tumor Molecular Alterations /Characteristics in Advanced / Metastatic Tumors.	II
NCT02465060	Molecular Analysis for Therapy Choice (MATCH).	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II
NCT04647838	A Phase II Study of Tepotinib in Patients With Solid Cancers Harboring c-MET Amplification or Exon 14 Mutation Who Progressed After Standard Treatment for Advanced/Metastatic Disease	II
No NCT ID	A Phase IB study of Crizotinib Either in Combination or as Single Agent in Pediatric Patients with ALK, ROS1 or MET Positive Malignancies	I
NCT04228406	Safety, Tolerability, Pharmacokinetic Characteristics, and Preliminary Efficacy Evaluation of the Selective c-MET Inhibitor GST-HG161 in Patients With Advanced or Metastatic Solid Tumors	I
NCT04169178	A Phase I Dose Finding/Expansion Study of HLX55, A Monoclonal Antibody Targeting Tyrosine-Protein Kinase MET (C-MET) in Patients With Advanced Solide Tumors Refactory to Standard Therapy	I
NCT02977364	Phase I Trial to Investigate Safety and Tolerability Profile and Pharmacokinetics of C-met Kinase Inhibitor HS-10241 in Subjects With Advanced Solid Tumours	I
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II

PTEN V133_C136del

NCT ID	Title	Phase
NCT03925350	A Phase II Study of Niraparib in Patients With Advanced Melanoma With Genetic Homologous Recombination (HR) Mutation / Alteration	II
NCT03994796	Genomically-Guided Treatment Trial in Brain Metastases	II
NCT03207347	A Phase II Trial of the PARP Inhibitor, Niraparib, in BAP1 and Other DNA Damage Response (DDR) Pathway Deficient Neoplasms (UF-STO-ETI-001)	II
NCT02286687	Phase II Study of the PARP Inhibitor Talazoparib in Advanced Cancer Patients With Somatic Alterations in BRCA1/2, Mutations/Deletions in Other BRCA Pathway Genes and Germline Mutation in BRCA1/2 (Not Breast or Ovarian Cancer)	II
NCT02401347	A Phase II Clinical Trial of the PARP Inhibitor Talazoparib in BRCA1 and BRCA2 Wild Type Patients With Advanced Triple Negative Breast Cancer and Homologous Recombination Deficiency or Advanced HER2 Negative Breast Cancer or Other Solid Tumors With a Mutation in Homologous Recombination Pathway Genes Talazoparib Beyond BRCA (TBB) Trial	II
NCT04317105	A Phase I/II Biomarker Driven Combination Trial of Copanlisib and Immune Checkpoint Inhibitors in Patients With Advanced Solid Tumors	I/II
NCT03842228	A Phase Ib Biomarker-Driven Combination Trial of Copanlisib, Olaparib, and MEDI4736 (Durvalumab) in Patients With Advanced Solid Tumors	I

Clinical Trials Summary (continued)

PTEN V133_C136del (continued)

NCT ID	Title	Phase
NCT04192981	A Phase I Study With Expansion Cohort of Concurrent GDC-0084 With Radiation Therapy for Patients With Solid Tumor Brain Metastases or Leptomeningeal Metastases Harboring PI3K Pathway Mutations	I
NCT03718091	A Phase II Study of M6620 (VX-970) in Selected Solid Tumors	II
NCT04591431	The Rome Trial From Histology to Target: the Road to Personalize Target Therapy and Immunotherapy	II
NCT02029001	A Two-period, Multicenter, Randomized, Open-label, Phase II Study Evaluating the Clinical Benefit of a Maintenance Treatment Targeting Tumor Molecular Alterations in Patients With Progressive Locally-advanced or Metastatic Solid Tumors MOST: My own specific treatment	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT03155620	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice) Screening Protocol	II
NCT03213678	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase II Subprotocol of LY3023414 in Patients With Solid Tumors	II
No NCT ID	Phase I/II Study of TAS-117 In Combination With TAS-120 In Patients With Advanced Solid Tumors	I/II
NCT03415659	A Phase I, Open-label, Single-center, Single/Multiple-dose, Dose-escalation/Dose-expansion Clinical Study on Tolerance and Pharmacokinetics of HWH340 Tablet in Patients With Advanced Solid Tumors	I
NCT03673787	Ice-CAP: A Phase I Trial of Ipatasertib in Combination With Atezolizumab in Patients With Advanced Solid Tumours With PI3K Pathway Hyperactivation	I/II
NCT03065062	Phase I Study of the CDK4/6 Inhibitor Palbociclib (PD-0332991) in Combination With the PI3K/mTOR Inhibitor Gedatolisib (PF-05212384) for Patients With Advanced Squamous Cell Lung, Pancreatic, Head & Neck and Other Solid Tumors	I

CDK6 amplification

NCT ID	Title	Phase
NCT03994796	Genomically-Guided Treatment Trial in Brain Metastases	II
NCT03310879	A Phase II Study of the CDK4/6 Inhibitor Abemaciclib in Patients With Solid Tumors Harboring Genetic Alterations in Genes Encoding D-type Cyclins or Amplification of CDK4 or CDK6	II
NCT02465060	Molecular Analysis for Therapy Choice (MATCH).	II
NCT03239015	Efficacy and Safety of Targeted Precision Therapy in Refractory Tumor With Druggable Molecular Event	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II
NCT04116541	MegaMOST - A Multicenter, Open-label, Biology Driven, Phase II Study Evaluating the Activity of Anti-cancer Treatments Targeting Tumor Molecular Alterations /Characteristics in Advanced / Metastatic Tumors.	II
NCT04591431	The Rome Trial From Histology to Target: the Road to Personalize Target Therapy and Immunotherapy	II
NCT03155620	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice) Screening Protocol	II

Clinical Trials Summary (continued)

CDK6 amplification (continued)

NCT ID	Title	Phase
NCT03526250	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice) - Phase 2 Subprotocol of Palbociclib in Patients With Tumors Harboring Activating Alterations in Cell Cycle Genes	II
NCT04247126	A Phase I Study of SY 5609, an Oral, Selective CDK7 Inhibitor, in Adult Patients With Select Advanced Solid Tumors	I

BRAF amplification

NCT ID	Title	Phase
NCT03905148	A Phase Ib, Open-Label, Dose-escalation and Expansion Study to Investigate the Safety, Pharmacokinetics and Antitumor Activities of a RAF Dimer Inhibitor BGB-283 in Combination With MEK Inhibitor PD-0325901 in Patients With Advanced or Refractory Solid Tumors	I/II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II
NCT04488003	A Two-Part, Phase II, Multi-center Study of the ERK Inhibitor Ulixertinib (BVD-523) for Patients With Advanced Malignancies Harboring MEK or Atypical BRAF Alterations	II
NCT03520075	A Phase I/II Study of the Safety, Pharmacokinetics, and Activity of ASTX029 in Subjects With Advanced Solid Tumors	I/II
NCT04305249	A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of ATG-017 Monotherapy in Patients With Advanced Solid Tumors and Hematological Malignancies	I
NCT04249843	A First-in-Human, Phase Ia/Ib, Open Label, Dose-Escalation and Expansion Study to Investigate the Safety, Pharmacokinetics, and Antitumor Activity of the RAF Dimer Inhibitor BGB-3245 in Patients With Advanced or Refractory Tumors	I
NCT02857270	A Phase I Study of an ERK1/2 Inhibitor (LY3214996) Administered Alone or in Combination With Other Agents in Advanced Cancer	I
NCT03634982	A Phase I, Open-Label, Multicenter, Dose-Escalation Study of RMC-4630 Monotherapy in Adult Participants with Relapsed/Refractory Solid Tumors	I

ATRX K425Q

NCT ID	Title	Phase
NCT03925350	A Phase II Study of Niraparib in Patients With Advanced Melanoma With Genetic Homologous Recombination (HR) Mutation / Alteration	II
NCT04187833	Phase II Trial of Nivolumab in Combination with Talazoparib in Patients with Unresectable or Metastatic Melanoma and Mutations in BRCA or BRCA-ness Genes.	II
NCT03207347	A Phase II Trial of the PARP Inhibitor, Niraparib, in BAP1 and Other DNA Damage Response (DDR) Pathway Deficient Neoplasms (UF-STO-ETI-001)	II

Clinical Trials Summary (continued)

ATRX K425Q (continued)

NCT ID	Title	Phase
NCT03842228	A Phase Ib Biomarker-Driven Combination Trial of Copanlisib, Olaparib, and MEDI4736 (Durvalumab) in Patients With Advanced Solid Tumors	I
NCT02278250	An Open-Label Study of the Safety, Tolerability, and Pharmacokinetic/Pharmacodynamic Profile of M4344 (Formerly VX-803) as a Single Agent and in Combination With Cytotoxic Chemotherapy in Participants With Advanced Solid Tumors	I
NCT03767075	Basket of Baskets: A Modular, Open-label, Phase II, Multicentre Study To Evaluate Targeted Agents in Molecularly Selected Populations With Advanced Solid Tumours	II
NCT03233204	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase 2 Subprotocol of Olaparib in Patients With Tumors Harboring Defects in DNA Damage Repair Genes	II
NCT04267939	An Open-label Phase Ib Study to Determine the Maximum Tolerated and/or Recommended Phase II Dose of the ATR Inhibitor BAY 1895344 in Combination With PARP Inhibitor Niraparib, in Patients With Recurrent Advanced Solid Tumors and Ovarian Cancer	I

Assay Information

Methodology, Scope, and Application:The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity:The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations:This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content:Thermo Fisher Scientific's Ion Torrent Oncomine Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of Oncomine Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in

'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:
AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:
ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:
AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:
AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGR, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSPO2, RSPO3, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

References

1. O'Leary et al. Reference sequence (RefSeq) database at NCBI: current status, taxonomic expansion, and functional annotation. Nucleic Acids Res. 2016 Jan 4;44(D1):D733-45. PMID: 26553804
2. Ryan et al. Snf2-family proteins: chromatin remodellers for any occasion. Curr Opin Chem Biol. 2011 Oct;15(5):649-56. PMID: 21862382
3. Heyer et al. Rad54: the Swiss Army knife of homologous recombination?. Nucleic Acids Res. 2006;34(15):4115-25. PMID: 16935872
4. Matsuda et al. Mutations in the RAD54 recombination gene in primary cancers. Oncogene. 1999 Jun 3;18(22):3427-30. PMID: 10362365
5. Abedalthagafi et al. The alternative lengthening of telomere phenotype is significantly associated with loss of ATRX expression in high-grade pediatric and adult astrocytomas: a multi-institutional study of 214 astrocytomas. Mod. Pathol. 2013 Nov;26(11):1425-32. PMID: 23765250
6. Clynes et al. ATRX dysfunction induces replication defects in primary mouse cells. PLoS ONE. 2014;9(3):e92915. PMID: 24651726
7. Tang et al. A novel transcription regulatory complex containing death domain-associated protein and the ATR-X syndrome protein. J. Biol. Chem. 2004 May 7;279(19):20369-77. PMID: 14990586
8. Xue et al. The ATRX syndrome protein forms a chromatin-remodeling complex with Daxx and localizes in promyelocytic leukemia nuclear bodies. Proc. Natl. Acad. Sci. U.S.A. 2003 Sep 16;100(19):10635-40. PMID: 12953102
9. Cerami et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. Cancer Discov. 2012 May;2(5):401-4. PMID: 22588877
10. Weinstein et al. The Cancer Genome Atlas Pan-Cancer analysis project. Nat. Genet. 2013 Oct;45(10):1113-20. PMID: 24071849
11. Pisapia. The Updated World Health Organization Glioma Classification: Cellular and Molecular Origins of Adult Infiltrating Gliomas. Arch. Pathol. Lab. Med. 2017 Dec;141(12):1633-1645. PMID: 29189064
12. Jiao et al. Frequent ATRX, CIC, FUBP1 and IDH1 mutations refine the classification of malignant gliomas. Oncotarget. 2012 Jul;3(7):709-22. PMID: 22869205
13. Cheng et al. Molecular testing for BRAF mutations to inform melanoma treatment decisions: a move toward precision medicine. Mod. Pathol. 2018 Jan;31(1):24-38. PMID: 29148538
14. Alrabadi et al. Detection of driver mutations in BRAF can aid in diagnosis and early treatment of dedifferentiated metastatic melanoma. Mod. Pathol. 2019 Mar;32(3):330-337. PMID: 30315274
15. Quan et al. The association between BRAF mutation class and clinical features in BRAF-mutant Chinese non-small cell lung cancer patients. Journal of Translational Medicine, 29 Aug 2019, 17(1):298. PMID: 31470866
16. Yao et al. Tumours with class 3 BRAF mutants are sensitive to the inhibition of activated RAS. Nature. 2017 Aug 10;548(7666):234-238. PMID: 28783719
17. Bracht et al. BRAF Mutations Classes I, II, and III in NSCLC Patients Included in the SLLIP Trial: The Need for a New Pre-Clinical Treatment Rationale. Cancers (Basel). 2019 Sep 17;11(9). PMID: 31533235
18. Cancer Genome Atlas Research Network. Integrated genomic characterization of papillary thyroid carcinoma. Cell. 2014 Oct 23;159(3):676-90. PMID: 25417114
19. Cancer Genome Atlas Network. Comprehensive molecular characterization of human colon and rectal cancer. Nature. 2012 Jul 18;487(7407):330-7. PMID: 22810696
20. Cancer Genome Atlas Research Network. Comprehensive molecular profiling of lung adenocarcinoma. Nature. 2014 Jul 31;511(7511):543-50. doi: 10.1038/nature13385. Epub 2014 Jul 9. PMID: 25079552
21. Wan et al. Mechanism of activation of the RAF-ERK signaling pathway by oncogenic mutations of B-RAF. Cell. 2004 Mar 19;116(6):855-67. PMID: 15035987
22. Tiacci et al. BRAF mutations in hairy-cell leukemia. N. Engl. J. Med. 2011 Jun 16;364(24):2305-15. PMID: 21663470
23. Diamond et al. Diverse and Targetable Kinase Alterations Drive Histiocytic Neoplasms. Cancer Discov. 2016 Feb;6(2):154-65. doi: 10.1158/2159-8290.CD-15-0913. Epub 2015 Nov 13. PMID: 26566875
24. Imielinski et al. Oncogenic and sorafenib-sensitive ARAF mutations in lung adenocarcinoma. J Clin Invest. 2014 Apr;124(4):1582-6. doi: 10.1172/JCI72763. Epub 2014 Feb 24. PMID: 24569458

References (continued)

25. Ciampi et al. Oncogenic AKAP9-BRAF fusion is a novel mechanism of MAPK pathway activation in thyroid cancer. J. Clin. Invest. 2005 Jan;115(1):94-101. PMID: 15630448
26. Palanisamy et al. Rearrangements of the RAF kinase pathway in prostate cancer, gastric cancer and melanoma. Nat. Med. 2010 Jul;16(7):793-8. PMID: 20526349
27. Jones et al. Tandem duplication producing a novel oncogenic BRAF fusion gene defines the majority of pilocytic astrocytomas. Cancer Res. 2008 Nov 1;68(21):8673-7. PMID: 18974108
28. Cin et al. Oncogenic FAM131B-BRAF fusion resulting from 7q34 deletion comprises an alternative mechanism of MAPK pathway activation in pilocytic astrocytoma. Acta Neuropathol. 2011 Jun;121(6):763-74. doi: 10.1007/s00401-011-0817-z. Epub 2011 Mar 20. PMID: 21424530
29. Ross et al. The distribution of BRAF gene fusions in solid tumors and response to targeted therapy. Int. J. Cancer. 2016 Feb 15;138(4):881-90. PMID: 26314551
30. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/202429s019lbl.pdf
31. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/202806s015lbl.pdf
32. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/210496s006lbl.pdf
33. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/204114s016lbl.pdf
34. https://www.accessdata.fda.gov/drugsatfda_docs/label/2019/210498s001lbl.pdf
35. https://www.accessdata.fda.gov/drugsatfda_docs/label/2018/206192s002lbl.pdf
36. https://dayonebio.com/news/day-one-announces-preliminary-phase-1-results-for-day101-in-pediatric-low-grade-glioma-receipt-of-breakthrough-therapy-designation-btd-and-new-phase-2-study/
37. https://www.cmedresearch.com/post/2020-07-bvd-rare-oncology-phase2-trial/
38. Kulkarni et al. BRAF Fusion as a Novel Mechanism of Acquired Resistance to Vemurafenib in BRAFV600E Mutant Melanoma. Clin. Cancer Res. 2017 Sep 15;23(18):5631-5638. PMID: 28539463
39. Johnson et al. Acquired BRAF inhibitor resistance: A multicenter meta-analysis of the spectrum and frequencies, clinical behaviour, and phenotypic associations of resistance mechanisms. Eur. J. Cancer. 2015 Dec;51(18):2792-9. PMID: 26608120
40. Nazarian et al. Melanomas acquire resistance to B-RAF(V600E) inhibition by RTK or N-RAS upregulation. Nature. 2010 Dec 16;468(7326):973-7. doi: 10.1038/nature09626. Epub 2010 Nov 24. PMID: 21107323
41. Rizos et al. BRAF inhibitor resistance mechanisms in metastatic melanoma: spectrum and clinical impact. Clin. Cancer Res. 2014 Apr 1;20(7):1965-77. PMID: 24463458
42. Shi et al. A novel AKT1 mutant amplifies an adaptive melanoma response to BRAF inhibition. Cancer Discov. 2014 Jan;4(1):69-79. PMID: 24265152
43. Van et al. The genetic landscape of clinical resistance to RAF inhibition in metastatic melanoma. Cancer Discov. 2014 Jan;4(1):94-109. doi: 10.1158/2159-8290.CD-13-0617. Epub 2013 Nov 21. PMID: 24265153
44. Villanueva et al. Concurrent MEK2 mutation and BRAF amplification confer resistance to BRAF and MEK inhibitors in melanoma. Cell Rep. 2013 Sep 26;4(6):1090-9. PMID: 24055054
45. Shi et al. Acquired resistance and clonal evolution in melanoma during BRAF inhibitor therapy. Cancer Discov. 2014 Jan;4(1):80-93. PMID: 24265155
46. Malumbres et al. Cell cycle, CDKs and cancer: a changing paradigm. Nat. Rev. Cancer. 2009 Mar;9(3):153-66. PMID: 19238148
47. Sherr et al. Targeting CDK4 and CDK6: From Discovery to Therapy. Cancer Discov. 2016 Apr;6(4):353-67. PMID: 26658964
48. Weinberg. The retinoblastoma protein and cell cycle control. Cell. 1995 May 5;81(3):323-30. PMID: 7736585
49. Cancer Genome Atlas Research Network. Comprehensive molecular characterization of gastric adenocarcinoma. Nature. 2014 Sep 11;513(7517):202-9. doi: 10.1038/nature13480. Epub 2014 Jul 23. PMID: 25079317
50. Cancer Genome Atlas Network. Comprehensive genomic characterization of head and neck squamous cell carcinomas. Nature. 2015 Jan 29;517(7536):576-82. PMID: 25631445

References (continued)

51. Peschard et al. A conserved DpYR motif in the juxtamembrane domain of the Met receptor family forms an atypical c-Cbl/Cbl-b tyrosine kinase binding domain binding site required for suppression of oncogenic activation. J. Biol. Chem. 2004 Jul 9;279(28):29565-71. PMID: 15123609
52. Peschard et al. Mutation of the c-Cbl TKB domain binding site on the Met receptor tyrosine kinase converts it into a transforming protein. Mol. Cell. 2001 Nov;8(5):995-1004. PMID: 11741535
53. Abella et al. Met/Hepatocyte growth factor receptor ubiquitination suppresses transformation and is required for Hrs phosphorylation. Mol. Cell. Biol. 2005 Nov;25(21):9632-45. PMID: 16227611
54. Sierra et al. c-MET as a potential therapeutic target and biomarker in cancer. Ther Adv Med Oncol. 2011 Nov;3(1 Suppl):S21-35. PMID: 22128285
55. Mo et al. Targeting MET in cancer therapy. Chronic Dis Transl Med. 2017 Sep;3(3):148-153. PMID: 29063069
56. Brennan et al. The somatic genomic landscape of glioblastoma. Cell. 2013 Oct 10;155(2):462-77. PMID: 24120142
57. Frampton et al. Activation of MET via diverse exon 14 splicing alterations occurs in multiple tumor types and confers clinical sensitivity to MET inhibitors. Cancer Discov. 2015 Aug;5(8):850-9. PMID: 25971938
58. Schrock et al. Characterization of 298 Patients with Lung Cancer Harboring MET Exon 14 Skipping Alterations. J Thorac Oncol. 2016 Sep;11(9):1493-502. PMID: 27343443
59. Pilotto et al. MET exon 14 juxtamembrane splicing mutations: clinical and therapeutical perspectives for cancer therapy. Ann Transl Med. 2017 Jan;5(1):2. doi: 10.21037/atm.2016.12.33. PMID: 28164087
60. Reungwetwattana et al. The race to target MET exon 14 skipping alterations in non-small cell lung cancer: The Why, the How, the Who, the Unknown, and the Inevitable. Lung Cancer. 2017 Jan;103:27-37. PMID: 28024693
61. Saffroy et al. MET exon 14 mutations as targets in routine molecular analysis of primary sarcomatoid carcinoma of the lung. Oncotarget. 2017 Jun 27;8(26):42428-42437. PMID: 28418914
62. Yeh et al. Activating MET kinase rearrangements in melanoma and Spitz tumours. Nat Commun. 2015 May 27;6:7174. doi: 10.1038/ncomms8174. PMID: 26013381
63. Bao et al. RNA-seq of 272 gliomas revealed a novel, recurrent PTPRZ1-MET fusion transcript in secondary glioblastomas. Genome Res. 2014 Nov;24(11):1765-73. PMID: 25135958
64. Sebastian et al. Recurrent MET fusion genes represent a drug target in pediatric glioblastoma. Nat. Med. 2016 Nov;22(11):1314-1320. PMID: 27748748
65. Zeng et al. c-Met gene amplification is associated with advanced stage colorectal cancer and liver metastases. Cancer Lett. 2008 Jul 8;265(2):258-69. PMID: 18395971
66. Tsugawa et al. Amplification of the c-met, c-erbB-2 and epidermal growth factor receptor gene in human gastric cancers: correlation to clinical features. Oncology. 1998 Sep-Oct;55(5):475-81. PMID: 9732228
67. Di et al. Overexpression and amplification of the met/HGF receptor gene during the progression of colorectal cancer. Clin. Cancer Res. 1995 Feb;1(2):147-54. PMID: 9815967
68. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/213591s000lbl.pdf
69. NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 2.2021]
70. https://www.emdgroup.com/en/news/tepotinib-breakthrough-therapy-designation-11-09-2019.html
71. Bean et al. MET amplification occurs with or without T790M mutations in EGFR mutant lung tumors with acquired resistance to gefitinib or erlotinib. Proc. Natl. Acad. Sci. U.S.A. 2007 Dec 26;104(52):20932-7. PMID: 18093943
72. Chen et al. Clinicopathologic and molecular features of epidermal growth factor receptor T790M mutation and c-MET amplification in tyrosine kinase inhibitor-resistant Chinese non-small cell lung cancer. Pathol Oncol Res. 2009 Dec;15(4):651-8. doi: 10.1007/s12253-009-9167-8. Epub 2009 Apr 21. PMID: 19381876
73. Suda et al. Reciprocal and complementary role of MET amplification and EGFR T790M mutation in acquired resistance to kinase inhibitors in lung cancer. Clin. Cancer Res. 2010 Nov 15;16(22):5489-98. PMID: 21062933
74. Zhang et al. Current mechanism of acquired resistance to epidermal growth factor receptor-tyrosine kinase inhibitors and updated therapy strategies in human nonsmall cell lung cancer. J Cancer Res Ther. 2016 Dec;12(Supplement):C131-C137. PMID: 28230005

References (continued)

75. Nguyen et al. Acquired resistance to epidermal growth factor receptor tyrosine kinase inhibitors in non-small-cell lung cancers dependent on the epidermal growth factor receptor pathway. Clin Lung Cancer. 2009 Jul;10(4):281-9. PMID: 19632948
76. Choueiri et al. Biomarker-Based Phase II Trial of Savolitinib in Patients With Advanced Papillary Renal Cell Cancer. J. Clin. Oncol. 2017 Sep 10;35(26):2993-3001. PMID: 28644771
77. Milella et al. PTEN: Multiple Functions in Human Malignant Tumors. Front Oncol. 2015 Feb 16;5:24. doi: 10.3389/fonc.2015.00024. eCollection 2015. PMID: 25763354
78. Song et al. The functions and regulation of the PTEN tumour suppressor. Nat. Rev. Mol. Cell Biol. 2012 Apr 4;13(5):283-96. PMID: 22473468
79. Chalhoub et al. PTEN and the PI3-kinase pathway in cancer. Annu Rev Pathol. 2009;4:127-50. PMID: 18767981
80. Leslie et al. Inherited PTEN mutations and the prediction of phenotype. Semin. Cell Dev. Biol. 2016 Apr;52:30-8. PMID: 26827793
81. Tan et al. Lifetime cancer risks in individuals with germline PTEN mutations. Clin. Cancer Res. 2012 Jan 15;18(2):400-7. PMID: 22252256
82. Dillon et al. Therapeutic targeting of cancers with loss of PTEN function. Curr Drug Targets. 2014 Jan;15(1):65-79. PMID: 24387334
83. Papa et al. Cancer-associated PTEN mutants act in a dominant-negative manner to suppress PTEN protein function. Cell. 2014 Apr 24;157(3):595-610. PMID: 24766807
84. Kato et al. Functional evaluation of p53 and PTEN gene mutations in gliomas. Clin. Cancer Res. 2000 Oct;6(10):3937-43. PMID: 11051241
85. Han et al. Functional evaluation of PTEN missense mutations using in vitro phosphoinositide phosphatase assay. Cancer Res. 2000 Jun 15;60(12):3147-51. PMID: 10866302
86. Mendes-Pereira et al. Synthetic lethal targeting of PTEN mutant cells with PARP inhibitors. EMBO Mol Med. 2009 Sep;1(6-7):315-22. PMID: 20049735
87. Bian et al. PTEN deficiency sensitizes endometrioid endometrial cancer to compound PARP-PI3K inhibition but not PARP inhibition as monotherapy. Oncogene. 2018 Jan 18;37(3):341-351. PMID: 28945226